

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 6 — Stoichiometry**Total Marks: 65****Time Allowed: 2h 30m****Version: 1****General Instructions:**

1. Write your name, roll number and section clearly on the answer sheet.
2. Section A is compulsory. Encircle the correct option on the question paper itself.
3. In Section B, attempt **all 11** questions. For each question, attempt either the main question **OR** its alternative.
4. In Section C, attempt **all 4** questions. For each question, attempt either the main question **OR** its alternative.
5. Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	The empirical formula of glucose (C ₆ H ₁₂ O ₆) is:	C ₆ H ₁₂ O ₆	CH ₂ O	C ₂ H ₄ O ₂	CHO	ⒶⒷⒸⒹ
2	The molecular mass of naphthalene (C ₁₀ H ₈) is:	120 amu	108 amu	128 amu	136 amu	ⒶⒷⒸⒹ
3	The formula mass of Mg(OH) ₂ is: (Mg=24, O=16, H=1)	40 amu	56 amu	58 amu	74 amu	ⒶⒷⒸⒹ
4	One mole of any substance contains how many representative particles?	6.022×10 ²²	6.022×10 ²³	6.022×10 ²⁴	6.022×10 ²⁰	ⒶⒷⒸⒹ
5	How many moles of molecules are there in 16 g of oxygen gas (O ₂)?	1 mol	0.5 mol	0.1 mol	2 mol	ⒶⒷⒸⒹ
6	The mass of 9.05 moles of ozone (O ₃) is: (O=16)	48 g	144 g	434.4 g	480 g	ⒶⒷⒸⒹ
7	How many atoms are present in 1.25 moles of zinc (Zn)?	6.022×10 ²³	7.53×10 ²³	4.82×10 ²³	1.25×10 ²³	ⒶⒷⒸⒹ
8	The formula of aluminium oxide (Al ³⁺ , O ²⁻) is:	AlO	Al ₂ O ₃	Al ₃ O ₂	AlO ₃	ⒶⒷⒸⒹ
9	Which compound has the same empirical and molecular formula?	H ₂ O ₂	C ₆ H ₆	H ₂ O	C ₆ H ₁₂ O ₆	ⒶⒷⒸⒹ
10	What is the mass of 4 moles of hydrogen gas H ₂ ? (H=1.008)	8.064 g	4.032 g	1 g	1.008 g	ⒶⒷⒸⒹ
11	In a chemical equation, balancing must be done by changing:	subscripts	symbols	coefficients	state symbols	ⒶⒷⒸⒹ
12	In an ionic equation, ions that appear on both sides and do not participate are called:	reactive ions	spectator ions	anions	cations	ⒶⒷⒸⒹ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	Define empirical formula and molecular formula. Give one example of each.	3	OR	What is a mole? State Avogadro's number and name three types of representative particles.	3
2(ii)	Calculate the molecular mass of glucose C ₆ H ₁₂ O ₆ . (C=12,	3	OR	Calculate the formula mass of CuSO ₄ ·5H ₂ O. (Cu=63.5, S=32,	3

	H=1.008, O=16)			O=16, H=1)	
2(iii)	Differentiate between gram atomic mass, gram molecular mass, and gram formula mass.	3	OR	Differentiate between molecular formula and empirical formula. Which two quantities are needed to find the molecular formula from the empirical formula?	3
2(iv)	Calculate the mass of 0.25 moles of CO ₂ . (C=12, O=16)	3	OR	A balloon is filled with 5 g of hydrogen gas (H ₂). Calculate the number of moles of H ₂ . (H=1.008)	3
2(v)	How many molecules are present in 0.5 moles of methane (CH ₄)?	3	OR	A sample of titanium contains 3.011×10^{23} Ti atoms. Calculate the number of moles of Ti. ($N_A = 6.022 \times 10^{23}$)	3
2(vi)	Write the chemical formula of: (a) sodium chloride (Na ⁺ , Cl ⁻) (b) magnesium nitride (Mg ²⁺ , N ³⁻) (c) potassium sulphate (K ⁺ , SO ₄ ²⁻)	3	OR	Write the structural formula and molecular formula of n-butane (C ₄ H ₁₀). Identify the structural formula of CH ₃ -CH ₂ -OH and write its molecular formula.	3
2(vii)	Balance the following chemical equation: CH _{4(g)} + O _{2(g)} → CO _{2(g)} + H ₂ O _(l)	3	OR	Balance the following chemical equation and add state symbols: Na _(s) + H ₂ O _(l) → NaOH _(aq) + H _{2(g)}	3
2(viii)	How many moles of molecules are in 3.011×10^{22} molecules of formaldehyde (CH ₂ O)? Also find its mass. (C=12, H=1, O=16)	3	OR	Calculate the number of atoms in 0.2 moles of aluminium (Al). ($N_A = 6.022 \times 10^{23}$)	3
2(ix)	Write the steps for writing the chemical formula of a binary ionic compound. Apply the steps to write the formula of aluminium oxide (Al ³⁺ , O ²⁻).	3	OR	Define ionic equation. Write the ionic equation for the reaction: HCl _(aq) + NaOH _(aq) → NaCl _(aq) + H ₂ O _(l)	3
2(x)	Calculate the number of moles in: (a) 15 g of NaCl (b) 40 g of sulphur (S). (Na=23, Cl=35.5, S=32)	3	OR	Calculate the mass of: (a) 1.2 moles of potassium (K) (b) 0.25 moles of steam (H ₂ O). (K=39, H=1.008, O=16)	3
2(xi)	Aspirin (C ₉ H ₈ O ₄) is used as a mild painkiller. (a) Write its empirical formula. (b) Calculate its molecular mass. (C=12, H=1, O=16)	3	OR	Caffeine is C ₈ H ₁₀ N ₄ O ₂ . (a) Write its empirical formula. (b) Calculate its molecular mass. (C=12, H=1, N=14, O=16)	3

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	OR Question	Marks
3(i)	(a) Define molecular mass and formula mass. Give one example of each with full calculation. [2] (b) Calculate the formula mass of KClO ₃ (potassium chlorate) and NaHCO ₃ (baking soda). (K=39, Cl=35.5, O=16, Na=23, H=1, C=12) [3]	5	OR	(a) Define empirical formula and molecular formula. Explain with benzene (C ₆ H ₆) and hydrogen peroxide (H ₂ O ₂) as examples. [2] (b) Write empirical formulas for: (i) Aspirin C ₉ H ₈ O ₄ (ii) Vinegar CH ₃ COOH (iii) TNT C ₇ H ₅ N ₃ O ₆ . Also calculate the molecular mass of each. [3]	5
	(a) Define mole and Avogadro's number. Explain the difference between gram atomic mass, gram molecular mass and gram formula			(a) Using the mole concept, calculate the number of moles in: (i) 2.4 g of He (ii) 250 mg of carbon (iii) 40 g of sulphur.	

3(ii)	mass. [3] (b) Calculate the number of molecules in: (i) 2.5 moles of CO ₂ (ii) 3.4 moles of NH ₃ . [2]	5	OR	(He=4, C=12, S=32) [3] (b) Calculate the mass of: (i) 75 moles of H ₂ (ii) 0.15 moles of H ₂ SO ₄ . (H=1.008, S=32, O=16) [2]	5
3(iii)	(a) Describe the steps for balancing a chemical equation. Why is it important to balance chemical equations? [2] (b) Balance the following equations and add correct state symbols: (i) NH ₃ → N ₂ + H ₂ (ii) Al + HCl → AlCl ₃ + H ₂ (iii) Fe + O ₂ → Fe ₂ O ₃ [3]	5	OR	(a) Define ionic equation. Explain the concept of spectator ions and state when they are removed. [2] (b) Transform the following into ionic equations: (i) Mg _(s) + H ₂ SO _{4(aq)} → MgSO _{4(aq)} + H _{2(g)} (ii) AgNO _{3(aq)} + NaCl _(aq) → AgCl _(s) + NaNO _{3(aq)} (iii) Zn _(s) + 2HCl _(aq) → ZnCl _{2(aq)} + H _{2(g)} [3]	5
3(iv)	(a) A 0.5 g sample of potassium (K) was added to water. [K=39, N _A =6.022×10 ²³] K _(s) + H ₂ O _(l) → KOH _(aq) + H _{2(g)} (i) Balance the equation. [1] (ii) Calculate the number of moles of K. [2] (iii) Calculate the number of K atoms in the sample. [2]	5	OR	(a) Potassium chlorate (KClO ₃) decomposes on heating to produce KCl and O ₂ . [K=39, Cl=35.5, O=16] (i) Write and balance the equation for this decomposition. [1] (ii) Calculate the formula mass of KClO ₃ . [2] (iii) How many moles of KClO ₃ are in 24.5 g? [2]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Chapter 6 — Stoichiometry (pp. 95-112)

— End of Question Paper —